

2 (a) (i) **Method 1**

Consider P and Q as 1 system.

$$W_Q - W_P = (m_Q + m_P) a$$

$$(1.00)(9.81) - (0.50)(9.81) = (1.00 + 0.50)a \quad [1]$$

$$a = 3.27 \text{ m s}^{-2} \quad [1]$$

Method 2

Consider each mass separately,

$$T - W_P = m_P a \quad \text{---(1)}$$

$$W_Q - T = m_Q a \quad \text{---(2)}$$

Solving the simultaneous equations (1) + (2),

$$(1.00)(9.81) - (0.50)(9.81) = (1.00 + 0.50)a \quad [1]$$

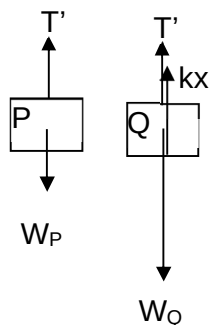
$$a = 3.27 \text{ m s}^{-2} \quad [1]$$

(ii) From equation (1) in 1(a)(i),

$$T = 0.50(9.81) + 0.50(3.27) \quad [1]$$

$$= 6.54 \text{ N (accept 3 or 4 s.f.)} \quad [1]$$

(iii)



After Block Q comes to a stop, the forces acting on P and Q are in equilibrium.

$$T' = W_P \quad \text{----- (3)}$$

$$T' + kx = W_Q \quad \text{----- (4)}$$

Substitute (3) into (4)

$$0.50(9.81) + 200x = 1.00(9.81) \quad [1]$$

$$x = 0.0245 \text{ m (3 s.f.)} \quad [1]$$

(iv) (All) initial KE is converted to EPE, which is (all) converted back to KE finally.

Hence, k.e. is conserved and collision is elastic. [1]

(b) (i) By conservation of momentum,

Initial momentum of bullet = final momentum of bullet and block

$$m_{\text{bullet}} \times 100 = (m_{\text{bullet}} + m_{\text{block}})v \quad [M1]$$

$$v = 0.990 \text{ ms}^{-1}$$

By conservation of energy,

Loss in k.e. of bullet and block = Gain in g.p.e of bullet and block

$$\frac{1}{2} (m_{\text{bullet}} + m_{\text{block}})v^2 = (m_{\text{bullet}} + m_{\text{block}})gh$$

$$h = \frac{1}{2} (0.990)^2 / (9.81) = 0.050 \text{ m} \quad [A1]$$

(ii) The collision is perfectly inelastic.

[1]